

KAUSTUV DEY

+91-8415020717

self.kaustuv@gmail.com

LinkedIn

GitHub

Leetcode

EDUCATION

• Dr. B R Ambedkar National Institute of Technology

M.Tech in IT (Data Analytics) | Current SGPA: 8.00 | CGPA: 7.80

Jalandhar, Punjab

Aug. 2024 – June 2026

• ICFAI University, Tripura

B.Tech in Computer Science & Engineering | CGPA: 8.76

Agartala, Tripura

Sept. 2020 – June 2024

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Golang, SQL, Bash/Shell, JavaScript
- **Tools/DevOps/Monitoring:** Git, Docker, Linux, CI/CD, GCP, JIRA, PagerDuty, QEMU
- **Data/ML:** Airflow, Vertex AI, BigQuery, Pandas, NumPy

EXPERIENCE

– Warner Bros. Discovery

Sept. 2025 - Present

Software Engineer Intern

Bangalore, India

- * Engineered architectural optimizations for cloud compute infrastructure, reducing pipeline runtime by 75% and driving \$8,000+/year in cost-effective resource scaling.
- * Provisioned and managed scalable virtual machine environments (GCP Workbenches), optimizing resource allocation and instance configurations for high-performance computing workloads
- * Orchestrated and deployed containerized workloads via Vertex AI Custom Jobs, automating execution environments to improve compute efficiency and system reliability.
- * Deployed 7+ production features and authored 80+ unit tests utilizing CI/CD principles, streamlining deployment processes and improving overall system stability.
- * Implemented robust infrastructure monitoring and alerting using PagerDuty and Slack integrations, significantly reducing on-call resolution times.
- * Troubleshoot and resolved 20+ critical production issues with hot-fixes across 5+ major microservices, providing advanced support to minimize system downtime.

– GAO Tek Inc.

Aug. 2022 - Apr. 2023

Tech Team Lead

Ontario, Canada

- * Optimized C code for 3 RFID tracking systems, improving accuracy by 85% and reducing errors by 40%.
- * Streamlined documentation for 40+ team members using Excel and Bitrix24, cutting reporting time by 60%.
- * Trained and mentored 6 junior developers in RFID and C programming, achieving 100% certification in 3 months.

– ZIROH Labs

Jan. 2022 - Mar. 2022

Software development Intern

Bangalore, India

- * Developed AES encryption for JSON data in a Maven-based project on Eclipse, ensuring robust security. Integrated REST APIs for secure communication and SQLite for efficient encrypted data storage, improving query performance by 20%.

PROJECTS

– On-Screen Text & Burnt-In Subtitle Detection Pipeline | Warner Bros. Discovery

- * Built pipeline to detect, extract, and classify on-screen text (graphics, subtitles, credits) from videos using **PaddleOCR + LLM post-processing**, evaluated on 70+ video-script pairs with 5,300+ annotated text blocks
- * Developed **BIS removal module** using Levenshtein + LCS string matching to improve **WER evaluation** accuracy by filtering burnt-in subtitles from ASR output while preserving legitimate dialogue.

– VisionTrack: AI-Powered CCTV Search & Geotagging System | Hackathon Project

- * Built an intelligent CCTV search and geotagging system leveraging a **Siamese model** for image similarity and **Google Maps API** for location mapping interacted using a simple website build on **Django**.
- * Integrated face, location, number plate recognition (using **YOLO** model) in both image and video format, to support Tripura State Police in criminal investigations and crime pattern analysis.

– My Kernel |

- * Developed a simple Unix-like Kernel in **C** and **Assembly** simulated on **QEMU**, featuring system calls, paging.
- * Built low-level **OS** components including a **Multiboot-compliant** bootloader, interrupt handling with a **Docker** build environment for portability.

ACHIEVEMENTS

– 2 research papers published on Quantum Steganography

- **GATE 2024 Qualified** in Computer Science and Information Technology with **91.36 percentile**. 2024
- **1st Runner-up** in Tripura Police Tech hackathon 2022-23 among **150 + teams** all over India. 2022-23
- **Start-Up IOT Project Funded** under New Generation Inovation Network (NGIN) by Govt. of Tripura.
- **Specialist** at Codeforces & **5 star** in C at Hackerrank